What is CSS?

CSS (Cascading Style Sheets) is a stylesheet language used to style and design HTML elements on a web page.

HTML defines the structure and content of a webpage, while CSS controls its appearance and layout.

Ways to Add CSS

CSS can be added to HTML in three ways:

1. Inline CSS

CSS is written directly inside the HTML element using the style attribute for applying a style to a single element.

style="css1: value; css2 : value;"

```
<h1 style="color: blue;">Hello World</h1>
```

2. Internal CSS

CSS is written inside the <style> </style> tag, usually within the <head></head> section for styling a single HTML page.

```
<head>
  <style>
    h1 {
      color: blue;
    }
  </style>
</head>
```

3. External CSS

CSS is written in a separate **.css** file and connected to HTML using the <link> tag.

Best approach for most real-world websites because the same CSS file can be used across multiple HTML pages.

— index.html (HTML FILE) —

```
<head>
  <link rel="stylesheet" href="style.css">
</head>
```

— style.css (CSS FILE) —

```
h1 {
  color: blue;
}
```

CSS Selectors

A CSS Selector is used to select the HTML element, that we want to style.

Syntax:

```
selector {
   property: value;
}
```

<u>Types of CSS Selectors</u>

1. Element Selector :

Element Selector selects all HTML elements of a specific type or tag.

```
ex.:    p {
            color: blue;
        }
```

2. ID Selector

ID Selector selects an HTML element using its unique id attribute using (#) symbol.

```
ex.:    #title {
            color: red;
        }
<h1  id="title">Apply red color only in this heading tag</h1>
```

3. Class Selector

Class Selector selects one or more HTML elements having the same class name using (.) symbol.

```
ex.:    .text {
            color: green;
        }
<h1  class="title">Apply red color only in this heading tag</h1>
<h3  class="title">Apply red color only in this heading tag</h3>
```

4. Universal Selector

Universal Selector selects all HTML elements on the page using (*) symbol.

```
ex.:    * {
            margin: 0;
            padding: 0;
        }
```

5. Grouping Selector

```
h1, h2, p {color: blue; }
```

SOME CSS PROPERTIES

1. CSS Colors

color : red **or** rgb(255, 255, 0) **or** #FEDA00; *(/* It is used for coloring the text only*/*

background-color : red **or** rgb(255, 255, 0) **or** #FEDA00; */* It is used for coloring the background/or any elements only*/*

border-color : red **or** rgb(255, 255, 0) **or** #FEDA00; */* It is used for coloring the outline of the box*/*

Color formats: (Name, RGB, RGBA, HEX)

2. CSS Background

Background properties control the color and image displayed (clipping mask image) behind an element.

Properties — values

----

background-color : red or #ff000 or rgb(255, 0, 255);

background-image : url(/images/bg-jpg);

background-repeat : no-repeat or no-repeat or repeat-x or repeat-y;

background-position : center or left or top or bottom or right or 50% 50%;

background-size : cover or contain or auto 100% 100%;

background-attachment : fixed or scroll (default value);

background: linear-gradient(rgba(0, 0, 0, 0.6), rgba(0, 0, 0, 0.6)), url('your-image.jpg') no-repeat center/cover;

3. CSS Text

Text properties control alignment decoration, transformation and spacing etc.

Properties — values

----

text-align : center or left or right or justify;

text-decoration : none or underline overline line-through;

text-transform : uppercase or lowercase or capitalize or none;

line-height : normal or 2px;

letter-spacing : normal or 2px;

word-spacing : normal or 2px;

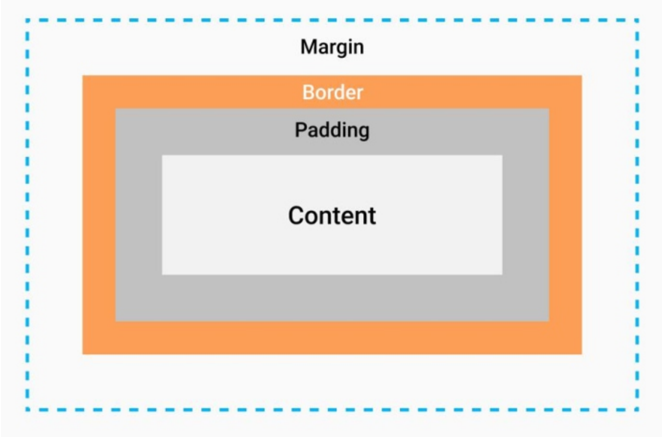
4. box-shadow

For apply a shadow effect on any box element like (any container)

box-shadow : 2px 2px 10px 5px #EAEAEB;

Box Model :

Every HTML element is treated as a rectangular box with these four parts - content, padding, border, and margin. *#4636#*#*# The Box Model describes how an element's total size is formed



5. CSS Border

Borders create a visible boundary around and element.

Properties — values

----

border-style :	solid or dashed or dotted or double or none;
border-width :	2px or thin or medium thick;
border-color :	red or #FED500 or rgb(......);
border-top :	2px solid red;
border-right :	2px solid red;
border-bottom	2px solid red;
border-left :	2px solid red;
border :	1px solid orange;

6. CSS Margin & Padding

Margin creates space outside of an element, while Padding creates space inside its element.

Properties — values

----

margin :	20px; */* Single value for all 4 side in clock wise - top, right, bottom, left)*
margin :	10px 20px; */* Two value for 2 by 2 side (first - top bottom , second right left)*
margin :	20px 6px 10px 20px; */* for all 4 side in clock wise - top, right, bottom, left)*
margin-top / right / bottom / left :	2px or thin or medium thick;
margin :	**auto or 0 auto;** ***/* This value is for center an element*/***
padding :	20px; */* Single value for all 4 side in clock wise - top, right, bottom, left)*
padding :	10px 20px; */* Two value for 2 by 2 side (first - top bottom , second right left)*
padding :	20px 6px 10px 20px; */* for all 4 side in clock wise - top, right, bottom, left)*
padding-top / right / bottom / left :	2px or thin or medium thick;

7. Display & Visibility :

Display controls how an element behaves and occupies space in the layout.

display : block or inline or inline-block or none;

Visibility: visibility controls whether an element is visible or hidden without changing its layout space.

visibility : visible or hidden;

8. Overflow controls content that goes outside an element's defined area.

overflow : visible or hidden or scroll or auto

overflow-x : visible or hidden or scroll or auto

overflow-y : visible or hidden or scroll or auto

9. CSS Position

Position properties control where an element is placed relative to its normal position or containing element.

position : static or relative or absolute or fixed or sticky

Note : relative will be treated as parent div and absolute will be treated as child of that relative div

top : 10px or 2rem or 10%

right : 10px or 2rem or 10%

bottom : 10px or 2rem or 10%

left : 10px or 2rem or 10%

z-index : 1 or 10 or 100

Note : z-index is responsible for layers arrangement (up & down any layer or element).

10. CSS Flexbox

Flexbox is a CSS layout system used to arrange elements inside a container in a row or column. It makes alignment, spacing and responsive layouts easier.

```
<div class="container">
   <div class="card">Item 1</div>
   <div class="card">Item 2</div>
   <div class="card">Item 3</div>
</div>
```

CSS Flexbox Explain

```
. container{
        display: flex;
        flex-direction: row   or   row-reverse   or   column   or   column-reverse;
        justify-content: value-below;
        align-items: value-below;
        gap: 20px;   or   row-gap: 20px;   or   column-gap: 30px;
        flex-wrap: wrap   or   nowrap   or   wrap-reverse;
}
```

flex-direction *decides the main direction in which flex items are arranged.*

row — Left to right
row-reverse — Right to left
column — Top to bottom
column-reverse — Bottom to top

---

justify-content *controls how flex items are distributed along the main axis.*

flex-start — Items start from the beginning
center — Items stay in the centre
flex-end — Items move to the end
space-between — Equal space between items
space-around — Space around each item
space-evenly — Equal space between and around items

---

align-items *controls the alignment of flex items along the cross axis.*

stretch — Items stretch by default
flex-start — Items move to the start
center — Items move to the centre
flex-end — Items move to the end

---

gap *is usually cleaner than adding margins to every child just to create spacing.*

---

flex-wrap — *By default, flex items try to stay on one line. flex-wrap controls whether they are allowed to move to another line.*

nowrap — All items stay on one line
wrap — Items move to the next line when required
wrap-reverse — Items wrap in the opposite cross-axis direction

---

Note — *This property is applied to the* ***child.***

.card {   flex-grow: 0   or   1   or   2;}

flex-grow controls how much a flex item can grow and take available extra space.